RUDRA CHAUHAN

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PROFESSIONAL SUMMARY

Final-year Computer Engineering student who builds full-stack, production-shaped AI systems end-to-end. Architected CortexGuard, a Dockerized AI Security Operations Center with real-time ML anomaly detection, LLM-driven threat intelligence, and an OWASP-compliant prompt-injection firewall. Comfortable across React/TypeScript frontends, FastAPI/Flask backends, and cloud deployment. Seeking Software Engineering, Full Stack, or AI Applications Engineering roles.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, Java, C, C++, SQL
- **Frontend:** React, TypeScript, Tailwind CSS, HTML5/CSS3, Vite
- **Backend & APIs:** FastAPI, Flask, REST API design, JWT authentication, WebSockets
- **AI, ML & Security:** scikit-learn (Isolation Forest anomaly detection), LLM integration (OpenAI/Ollama), Prompt Engineering, Prompt-Injection Defense (OWASP LLM Top 10), MITRE ATT&CK mapping, Deepfake/Media Forensics
- **Databases & Cloud:** PostgreSQL, Firebase (Auth, Firestore, Storage), Nginx, Vercel, Render
- **Tools:** Git, GitHub, Swagger/OpenAPI, VS Code, Postman

PROJECTS

CortexGuard — AI-Powered Autonomous Security Operations Center | *FastAPI · React · PostgreSQL · Docker · scikit-learn* GitHub

- Architected a full-stack SOC platform (React + FastAPI + PostgreSQL, Dockerized with Nginx reverse proxy) that ingests raw SSH/Apache/Nginx/firewall logs and pipelines them through parsing, detection, and AI triage in real time.
- Engineered a dual detection layer: an 8-rule sliding-window engine for known attack signatures plus an Isolation Forest anomaly model (10-feature vector, incremental training, joblib persistence) for unknown threats.
- Designed a 3-layer Prompt Firewall (16 regex patterns + hard-blocks, output-schema validation, full audit trail) to protect the LLM threat-analysis pipeline from adversarial log-injected prompts, aligned to OWASP LLM01.
- Integrated LLM-based alert analysis (OpenAI/Ollama, switchable) that maps every alert to one of 18 MITRE ATT&CK techniques with a recommended remediation action, validated for schema compliance and credential-leak prevention.
- Implemented JWT/bcrypt authentication with role-based access control, per-IP sliding-window rate limiting, and a global error handler that suppresses stack traces in production.

FrameGuard — Deepfake & AI-Generated Media Detection | *Flask · Python · Multimodal ML* GitHub · Live Demo

- Built an agent-based Flask web application that analyzes images, video frames, and audio to flag deepfake or AI-generated content through a multimodal forensic scoring pipeline.
- Designed a production-style split between a lightweight always-on cloud service (Render) for validation and a compute-heavy local pipeline handling frame extraction, audio analysis, and full forensic report generation.
- Presented system architecture, threat model, and multimodal decision-fusion strategy to technical judges as part of the project deliverables.

CVision — Intelligent Resume Screening Platform | *React · TypeScript · Firebase · Vite* GitHub

- Developed a recruiter-facing web app (React, TypeScript, Tailwind, Vite) for uploading, storing, and organizing candidate resumes against job postings, backed entirely by Firebase Auth, Firestore, and Storage — no custom backend server.
- Implemented a structured, rule-based resume-parsing pipeline that extracts skills and experience fields into a consistent recruiter-facing summary, with security rules scoping data access to each recruiter's own records.
- Isolated all Firebase read/write logic in a dedicated services layer to keep UI components presentation-only and the codebase maintainable as features grow.

EDUCATION

B.E. Computer Engineering — C.K. Pithawala College of Engineering & Technology (GTU), Surat
Jul 2023 – 2027 (Expected)
CGPA: 8.22/10 | Currently in 7th Semester

ACHIEVEMENTS & CERTIFICATIONS

- **NPTEL — Data Mining, IIT Kharagpur** (8-week certification, Jan–Mar 2026, Roll No. NPTEL26CS58S669100285)
- **Winner Techmanthan 2026 Hackathon — Techno NJR Institute of Technology, Udaipur** (Team Algoholics, Apr 2026)
- **Climate Data Hackathon — Delhi 2026, Climate Collective** (Finalist)
- **Remarkskill Education — Ethical Hacking Summer Internship** (Jun–Jul 2025); delivered a vulnerability assessment report as the internship capstone
- **YBI Foundation — Python Programming for AI & Data Science** (2-week virtual internship, completed Jul 2026)
- **Hack & Forge — Data Science Summit 2026, BIT Mesra Ranchi** (participant, via Unstop)